

JavaScript Syllabus for Full-Stack Development

1. Core JavaScript Fundamentals

- Variables and Data Types: var, let, const; strings, numbers, arrays, objects
- Operators and Expressions: Arithmetic, logical, comparison
- Functions: Declarations, expressions, arrow functions, higher-order functions
- Control Structures: if, else, switch, loops (for, while)
- Error Handling: try, catch, finally

2. Object-Oriented Programming in JavaScript

- Objects and Classes: Creating objects, using constructors, prototypes
- Inheritance and Encapsulation: Understanding this, class inheritance, encapsulation

3. Asynchronous JavaScript

- Callbacks and Promises: Understanding asynchronous behavior, chaining promises
- Async/Await: Simplifying asynchronous code with async/await
- Event Loop: Understanding the event loop and its impact on asynchronous operations

4. Front-End JavaScript

- DOM Manipulation: Selecting elements, modifying attributes, event handling
- Event Handling: Handling user interactions, form submissions
- ES6+ Features: Arrow functions, template literals, destructuring, modules (import/export)
- Responsive Design with JavaScript: Media queries, dynamic styling

5. Front-End Frameworks/Libraries

- Introduction to React.js: Components, state, props, lifecycle methods
- Single Page Applications (SPAs): React Router for client-side routing
- State Management: Introduction to Redux for managing global state

6. Back-End JavaScript (Node.js)

- Node.js Basics: Introduction to Node.js, understanding the event-driven architecture
- Building a Simple Server: Using Node.js and Express.js to create a RESTful API
 - Routing and Middleware: Handling different HTTP requests, using middleware for tasks like logging, authentication

7. Database Integration

- Working with MongoDB: Basics of NoSQL databases, CRUD operations using Mongoose
- SQL Basics: Introduction to relational databases, using JavaScript to interact with SQL databases (e.g., using Sequelize)
- Data Validation and Sanitization: Ensuring data integrity and security

8. Authentication and Security

- User Authentication: Implementing JWT (JSON Web Tokens) for secure user authentication
- Authorization: Role-based access control (RBAC), protecting routes
- Security Best Practices: Handling sensitive data, preventing SQL injection and XSS attacks

9. Real-Time Communication

- WebSockets with Socket.io: Building real-time applications like chat apps, notifications
- Polling vs. WebSockets: Understanding different methods for real-time data fetching

10. Testing and Debugging

- Unit Testing: Writing and running tests using Jest or Mocha
- Debugging Tools: Using Chrome DevTools for debugging both front-end and back-end JavaScript code
- Test-Driven Development (TDD): Writing tests before code for better reliability

11. DevOps and Deployment

- Package Management with npm: Installing and managing dependencies
- Build Tools: Using Webpack and Babel for compiling and bundling code
- CI/CD Pipelines: Basics of continuous integration/continuous deployment
- Deploying Full-Stack Applications: Deploying applications on platforms like Heroku, Netlify, or AWS

12. Capstone Project

- Build a Full-Stack Application: Combine all the concepts to build a real-world application (e.g., a task manager, e-commerce site)
- Version Control with Git: Using Git and GitHub for version control and collaboration
- Final Presentation: Showcase your project, focusing on both the front-end and back-end components